

# ANKIT VATS

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| Key Skills               | Exp        |
|--------------------------|------------|
| linux kernel development | 4.10 Years |
| X86/ARM                  | 3.8 Years  |

## OBJECTIVE

Software Engineer with 4.10 years of experience in Embedded-Field , seeking full-time developer roles.

## SKILLS

|                         |                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------|
| <b>Technical Skills</b> | C, C++, Inter-Process Communication, Firmware-Development, Schematics-verification, Embedded Linux |
| <b>Protocols</b>        | UART, I2C, SPI.                                                                                    |
| <b>Tools</b>            | EZ-USB Suite, LPCXpresso IDE, Flash-Magic, USB-Analyzer, Wireshark                                 |
| <b>Editor</b>           | Visual-Studio, VIM-Terminal                                                                        |

## EXPERIENCE

### Senior Research and Development Engineer

Nov 2020 - Present

Logic Fruit Technology

Gurugram

- Developed the Firmware for LPC1788 ARM Cortex-M3 32-bit MCU which configure the multiple Pins and peripheral of the platform-Board.
- Developed the Firmware for EZ-USB FX3, a Cypress's USB 3.0 peripheral controller. FX3 integrates the USB 3.0 and USB 2.0 physical layers (PHYs) along with a 32-bit ARM926EJ-S microprocessor for powerful data processing and for building custom USB SuperSpeed applications
- Design API's for seamless integration between x86 Application and custom Hardware. These API's provide the multiple-hardware configurations and Hardware-status and data-transfer.
- Collaborated with cross-functional teams to gather requirements, conduct code reviews, and troubleshoot issues, ensuring timely delivery of high-quality software solutions.
- Currently developing the Firmware for NXP K20 MCU and State-Machine based implementation.

### Trainee

Jun 2019 - Jun 2020

EmbLogic Research and Competency Development Labs

Noida

- Linux System Programming in which cover Inter-Process Communication (IPC), synchronization techniques, threads and get the comprehensive understanding of system-level programming in Linux.
- Character Device Driver Development in which covers the fundamentals of device driver development, focusing on character devices that include device registration, and communication between user space and kernel space.

## PROJECTS

### Current-Project:

Baseboard Management Controller(BMC) : It provides out-of-band management and monitoring of the system hardware like Monitoring Hardware, Remote Management, System Alerts.

**Corvette.** In this Project, a platform-board having the NXP-LPC1788 MCU and five PCIe port, LEDs, FAN, UART and EEPROM on I2C as peripherals. And two more hardware, Gen5/Gen6 PCIe Analyzer and Exerciser connected over

the PCIe port on platform. Exerciser is used to generate the GEN5/GEN6 PCIe packets and analyze by the Analyzer based PCIe Gen5/GEN6 protocol standards. And these two hardware provides the USB-interface to the x86 machine. Develop the corresponding x86 .dll library for Window Analyzer/Exerciser application to provide the high-data transfer on USB3.0 standard for various hardware configurations.

**Cytox.** Build a project in which server is running on Xilinx FPGA and Client is on x86 machine, connected through ethernet interface. Write the client code using Sockets that able to receive the data at speed of 2.4 Mbps on single channel.

**ARINC818.** Build a project that convert the HDMI video input to ARINC818 video Output based upon ARINC818 protocol. Here upgrade the USB2.0 interface to USB3.0 by using FX3-USB peripheral Controller. And also develop the GUI for Window and for Linux by providing the APIs.

## **EDUCATION**

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**Bachelor of Technology (ECE), Maharshi Dayanand University** 2017-2020

**Goverment Polytechnic (ECE), Sonipat** 2015 - 2017

